Reality Alignment Module (SRAM) defines the governance conditions under

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™
Architecture Family: SIMULOS® — Simulation Governance Architecture
Parent Standard: Simulation-to-Reality Correlation Standard (SRCS)
Operational Layer: Simulation-to-Reality Correlation Layer
Category: Governance & Enforcement
Subcategory: Simulation Governance
Type: Parent Standard Module
Governed Space: Reality Alignment
Version: 1.0
Status: Canonical · Open Module
Effective Date: 19 August 2026
Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · RIS™
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Governed Space

Reality Alignment

Canonical Definition

Reality Alignment Module (SRAM) defines the governance conditions under which material Simulation-Reality Deviations are translated into controlled alignment requirements, priorities, actions, boundaries, and reassessment pathways so that a simulation may be brought into closer correspondence with the Operational Reality it is intended to represent without erasing legitimate abstraction, uncertainty, simplification, or known differences.

SRAM governs the transition from:

Known Simulation-Reality Deviation

to:

Governed Reality Alignment.

Its purpose is not to force simulation and reality into artificial equivalence.

Its purpose is to determine:

which deviations require alignment, what should be aligned, how alignment should occur, what should legitimately remain different, and whether the resulting simulation-reality relationship has materially improved.

Operational Role

SRAM is the fourth internal governance space of the Simulation-to-Reality Correlation Standard (SRCS).

The internal progression is:

SRCM — Simulation-Reality Comparison Module

→ establishes what may legitimately be compared.

SCCM — Simulation Correlation Criteria Module

→ establishes the criteria governing correspondence.

SDMM — Simulation-Reality Deviation Module

→ identifies and structures material differences.

SRAM — Reality Alignment Module

→ determines how identified deviations should affect the simulation-reality relationship.

SCUM — Correlation Uncertainty Module

→ governs uncertainty remaining after correlation and alignment.

SRAM therefore asks:

Given what we now know about the difference between simulation and reality, what should change, what should remain unchanged, and how should the new alignment state be established?

Module Operational Space

SRAM governs: alignment requirement identification, alignment necessity, alignment eligibility, alignment prioritization, alignment scope, alignment target definition, alignment depth, alignment pathway selection, model alignment, environment alignment, parameter alignment, assumption alignment, input alignment, scenario alignment, behavioral alignment, temporal alignment, spatial alignment, interaction alignment, boundary alignment, condition-specific alignment, subsystem alignment, alignment dependencies, alignment sequencing, alignment authority, alignment controls, alignment implementation, alignment traceability, alignment verification, post-alignment comparison, residual deviation, alignment effectiveness, alignment limitations, alignment conflict, over-alignment prevention, alignment rollback, alignment status, and alignment documentation.

Module Function

SDMM establishes that a difference exists.

SRAM determines what should happen because of that difference.

It prevents organizations from assuming that every deviation should automatically result in modification of the simulation.

A deviation may instead require:

Correction

Calibration

Refinement

Boundary Restriction

Scenario Expansion

Parameter Adjustment

Representation Change

Assumption Revision

Environmental Adjustment

Additional Reality Observation

or:

No Alignment Action

where the difference is intentional, immaterial, unavoidable, or legitimately outside the simulation purpose.

Deviation Does Not Automatically Require Alignment

SIMULOS® establishes:

Observed deviation creates an alignment question, not an automatic alignment instruction.

A simulation is necessarily a representation.

Some differences from reality may be: intentional, methodologically necessary, computationally justified, immaterial, outside scope, purpose-specific, or inherent to abstraction.

SRAM therefore distinguishes:

Deviation

from:

Alignment Requirement.

Alignment Requirement

An Alignment Requirement exists where an identified Simulation-Reality Deviation materially weakens the simulation's intended correspondence with reality and sufficient basis exists to require a governed response.

Conceptually:

Governed Deviation

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Materiality Assessment

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Simulation Purpose & Scope

↓

Correlation Criteria

↓

Alignment Necessity

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Alignment Requirement Alignment Necessity

For every material deviation, SRAM should ask:

Does this difference actually require the simulation to change?

Possible outcomes may include:

Alignment Required

Alignment Conditionally Required

Alignment Recommended

Alignment Not Required

Alignment Not Currently Feasible

or:

Alignment Necessity Undetermined.

Alignment Eligibility

Not every observed deviation is sufficiently understood to justify modification.

Alignment Eligibility considers whether: the deviation is sufficiently established, the comparison basis is reliable, Operational Reality is sufficiently understood, the probable alignment target is identifiable, relevant dependencies are known, and modification would not introduce greater uncertainty.

Do Not Calibrate to Uncertain Reality

A critical SRAM principle is:

A simulation should not be modified merely to fit a reality observation whose own reliability is materially uncertain.

Where the reality basis is insufficient, additional Operational Reality evidence may be required before alignment.

Alignment Target

An Alignment Target identifies the specific simulation element or relationship requiring modification.

Targets may include: Reality Representation, assumption, model component, environment, parameter, input transformation, temporal behavior, scenario condition, interaction rule, subsystem relationship, response function, boundary condition, or another simulation element.

Alignment Target ≠ Deviation Location

The place where a deviation becomes visible may not be the place where alignment should occur.

For example:

Trajectory Deviation

may originate from:

Sensor Representation Error

rather than the trajectory model itself.

Therefore:

Observed Deviation Location ≠ Proven Alignment Target.

Alignment Attribution

Before modifying a simulation, SRAM should determine whether sufficient evidence connects the deviation to the proposed alignment target.

Where attribution remains uncertain:

Alignment Target Undetermined

should remain legitimate.

Alignment Priority

Not all deviations require equal attention.

Alignment priority may depend upon:

Materiality

Consequence

Frequency

Operational Relevance

Scenario Importance

Deviation Propagation

Reliance

Correctability

and:

Uncertainty.

Critical Alignment

A deviation may receive Critical Alignment priority where simulation-reality divergence materially affects: safety, mission success, autonomous behavior, critical infrastructure, high-consequence decisions, system boundaries, or downstream reliance.

Alignment Scope

Alignment should be bounded.

Possible scopes include:

Local Alignment

Subsystem Alignment

Scenario-Specific Alignment

Environment-Specific Alignment

Parameter-Range Alignment

Behavioral Alignment

Whole-Model Alignment

or:

Multi-Layer Alignment.

Minimum Necessary Alignment Principle

SIMULOS® establishes:

Alignment should modify no more of the simulation than is justified by the governed deviation basis.

This reduces the risk that correction of one region unintentionally damages correlation elsewhere.

Local Alignment

A localized deviation may require only a localized change.

For example:

incorrect low-temperature battery behavior

should not automatically require redesign of unrelated simulation components.

Condition-Specific Alignment

A simulation may require alignment only under defined conditions.

For example:

Nominal Conditions → Existing Representation Retained

High Radiation → Updated Representation

Low Power + High Radiation → Additional Conditional Model

This allows the simulation to preserve valid existing behavior while improving deficient regions.

Model Alignment

Model Alignment modifies the simulation model where evidence demonstrates that model structure or behavior contributes materially to Simulation-Reality Deviation.

Environment Alignment

Environment Alignment modifies the simulated environment where reality observations demonstrate that the environment does not sufficiently represent relevant Operational Reality.

Parameter Alignment

Parameter Alignment adjusts parameter values or parameter relationships where the governing basis supports such adjustment. Parameter Fitting ≠ Reality Alignment

A simulation can be tuned until historical observations match closely while still representing reality poorly outside the fitted region.

SIMULOS® therefore establishes:

Parameter fit alone does not establish Reality Alignment.

Assumption Alignment

Where an assumption materially contributes to deviation, SRAM may require reassessment of that assumption.

The assumption itself remains governed under SAS — Simulation Assumption Standard.

SRAM governs the correlation-driven requirement to revisit it.

Input Alignment

Where deviation results from input representation or transformation, SRAM may route the alignment requirement toward the relevant SIDS governance space.

Scenario Alignment

Reality observations may reveal operational conditions missing from the Scenario Space.

In such cases, alignment may require:

new scenario identification,

new variations,

new edge cases,

new stress conditions,

or:

expanded scenario coverage.

The relevant scenario design remains governed by SSS — Simulation Scenario Standard.

Behavioral Alignment

Behavioral Alignment concerns simulation behavior that differs materially from observed Operational Reality.

This may involve: autonomous decisions, agent responses, human-system interaction, recovery behavior, coordination, escalation, adaptation, or other dynamic behavior.

Temporal Alignment

Temporal Alignment addresses material differences in: response latency, event timing, sequencing, duration, recovery time, synchronization, and transition speed.

Spatial Alignment

Spatial Alignment may address differences in: trajectory, position, geometry, movement, distribution, coverage, or spatial interaction.

Interaction Alignment

Complex simulations may accurately represent individual components while poorly representing interactions among them.

SRAM therefore permits alignment of:

relationships

rather than merely individual objects.

Alignment Dependencies

A proposed alignment may affect multiple simulation elements.

For example:

Environment Alignment

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Sensor Response Changes

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Autonomous Decision Changes

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Trajectory Changes

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Output Distribution Changes

Alignment dependencies should therefore be mapped before material modification where practical.

Alignment Sequencing

Where multiple deviations share dependencies, alignment actions should be sequenced so that upstream corrections can be evaluated before unnecessary downstream modifications are made.

Conceptually:

Probable Root Alignment

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Re-Execution

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Recomparison

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Residual Deviation

↓

Additional Alignment if Required No Cascading Correction Without Reassessment

SIMULOS® establishes:

A correction should not automatically trigger an uncontrolled chain of additional corrections without renewed simulation-reality assessment.

Alignment Authority

Material alignment actions should have identifiable authority where changes can affect: safety conclusions, mission models, autonomous behavior, regulatory evidence, operational reliance, validation, or high-consequence decisions.

Alignment Implementation

Implementation may include: model modification, environment modification, parameter update, assumption revision, scenario expansion, input change, boundary change, resolution increase, interaction-model change, or another controlled simulation modification.

Alignment Traceability

Every material alignment should remain traceable through:

Deviation ID

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Alignment Requirement

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Alignment Target

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Alignment Action

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Simulation Version Change

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Re-Execution

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Post-Alignment Comparison

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Alignment Outcome Alignment Does Not End With Modification

A simulation is not considered aligned merely because someone changed it.

The change must be reassessed against Operational Reality.

Therefore:

Modification ≠ Alignment.

Post-Alignment Comparison

After material alignment, the affected simulation region should be compared again against the relevant Operational Reality basis.

This creates:

Pre-Alignment Deviation

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Alignment Action

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Post-Alignment Comparison

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Residual Deviation Alignment Effectiveness

Alignment Effectiveness asks:

Did the governed modification materially improve the intended Simulation-Reality Correlation?

Possible outcomes may include:

Alignment Effective

Alignment Partially Effective

Alignment Conditionally Effective

Alignment Ineffective

Alignment Degraded Other Regions

or:

Alignment Effectiveness Undetermined.

Residual Deviation

Alignment does not require elimination of all difference.

A Residual Deviation is the difference remaining after alignment.

Residual deviations may be: acceptable, conditional, material, unresolved, or uncertain.

They must remain visible.

No Zero-Deviation Requirement

SIMULOS® does not establish:

Simulation = Reality

as the goal.

The appropriate goal is:

sufficiently governed correspondence for the defined simulation purpose.
Over-Alignment

Over-Alignment occurs when a simulation is modified beyond what the evidence and intended purpose justify.

This can result in: overfitting, loss of generalizability, hidden assumptions, reduced scenario performance, artificial historical fit, or degraded representation elsewhere.

No Reality Overfitting Principle

SIMULOS® establishes:

A simulation should not become so fitted to observed historical reality that it loses legitimate capability to represent unobserved but materially plausible conditions.

This is especially important for: rare-event simulation, safety analysis, mission simulation, autonomous systems, climate and environmental simulation, and stress testing.

Alignment Conflict

An Alignment Conflict exists where improving correlation in one region materially worsens correlation in another.

For example:

Alignment improves nominal behavior

but:

degrades edge-case behavior.

Such conflict must remain explicit.

Multi-Objective Alignment

Where multiple correlation objectives exist, alignment may require balancing: nominal accuracy, edge-case representation, temporal behavior, physical behavior, system interaction, safety-critical behavior, and computational feasibility.

Alignment Rollback

Where an alignment action materially worsens the simulation-reality relationship, the implementation should support rollback or equivalent restoration where technically feasible.

Alignment Versioning

Material alignment should create or identify a new simulation version where the change affects correlation behavior.

Conceptually:

Simulation v2.4

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Alignment A-017

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Simulation v2.5

This preserves the relationship between simulation versions and reality-correlation history.

Reality Alignment Status

A simulation or defined simulation region may receive a status such as:

Aligned

Conditionally Aligned

Partially Aligned

Alignment Required

Alignment In Progress

Alignment Unresolved

or:

Alignment Status Undetermined.

These states should always remain bounded to the applicable correlation scope.
Aligned ≠ Reality Equivalent

A critical SIMULOS® rule is:

Aligned does not mean equivalent to reality.

It means that within the defined comparison basis, criteria, conditions, and known uncertainty, the simulation-reality relationship satisfies the applicable alignment requirements.

Alignment Across Simulation Versions

SRAM may preserve:

Version

→ Deviation Profile

→ Alignment Actions

→ Post-Alignment Correlation

This creates a historical record of whether simulation evolution actually improves correspondence with reality.

Alignment Regression

A new simulation version may solve one deviation while reintroducing an older one.

SRAM should therefore support detection of:

Reality Alignment Regression.

Alignment Learning

Repeated alignment cycles may reveal persistent structural weaknesses.

For example:

Repeated environment-related deviations

may indicate that the underlying Reality Representation requires deeper revision rather than continued parameter adjustment.

Escalation to Upstream SIMULOS® Standards

SRAM does not silently take ownership of every simulation component it affects.

Instead, it can route required changes back to the appropriate Parent Standard.

For example:

Representation problem

→ RRS

Assumption problem

→ SAS

Model / Environment problem

→ SMES

Input problem

→ SIDS

Scenario problem

→ SSS

Execution problem

→ SES

This preserves architectural separation.

Alignment Re-Entry

After an upstream simulation element changes, affected downstream SIMULOS® governance spaces may need reassessment.

Conceptually:

Reality Deviation

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Alignment Requirement

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Upstream Simulation Change

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Affected Lifecycle Re-Entry

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Re-Execution

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New Output

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New Simulation-Reality Comparison

This makes SIMULOS® capable of iterative simulation improvement without collapsing governance spaces into one correction process.

Relationship to SRCM

SRCM supplies the governed comparison basis.

SRAM depends upon that basis when determining whether alignment materially improves correspondence.

Relationship to SCCM

SCCM provides the criteria defining relevant correspondence.

SRAM should not optimize alignment against an undeclared or changing correlation criterion.

Relationship to SDMM

SDMM produces the governed deviation basis.

SRAM consumes that basis and determines the appropriate alignment response.

Therefore:

Deviation Identification precedes Alignment Action.

Relationship to SCUM

Even after successful alignment, uncertainty remains.

SCUM governs uncertainty in the resulting simulation-to-reality correlation relationship.

Therefore:

Alignment does not eliminate Correlation Uncertainty.

Relationship to ORA™

ORA™ governs Operational Reality itself.

SRAM does not redefine reality to make the simulation appear accurate.

SIMULOS® uses sufficiently governed reality information to govern the alignment relationship.

Relationship to VALIDOS®

Successful Reality Alignment does not automatically establish validation.

VALIDOS® may use simulation-reality correlation and alignment evidence as part of a broader Validation Determination.

Therefore:

Reality Alignment ≠ Validation.

Relationship to INTEGROS®

INTEGROS® may govern integrity properties of alignment evidence, configuration records, measurements, and resulting artifacts.

SRAM owns the simulation-specific Reality Alignment relationship.

Autonomous Systems Application

Autonomous systems may behave differently in simulation and Operational Reality because of: sensor imperfections, timing, physical interactions, human behavior, communication conditions, environment complexity, agent interactions, hardware limitations, or adaptive behavior.

SRAM can identify which deviations require actual simulation changes rather than indiscriminate tuning.

For example:

Simulated autonomous recovery: 1.8 seconds

Observed operational recovery: 3.1 seconds

The difference alone does not tell engineers what to modify.

SRAM requires the deviation to be traced sufficiently before an Alignment Target is established.

Space and Mission Simulation Application

Space simulation creates an especially important Reality Alignment problem because the most consequential mission conditions may have limited direct real-world observations.

A spacecraft simulation may be calibrated against: ground testing, hardware-in-the-loop testing, historical missions, telemetry, subsystem experiments, environmental observations, and limited operational flight data.

SRAM prevents the available observations from being treated as complete reality.

Alignment may improve correspondence for known conditions while preserving uncertainty for: unobserved radiation combinations, rare subsystem interactions, unknown surface conditions, extreme thermal states, long-duration degradation, communication anomalies, and autonomous recovery behavior.

The goal is therefore not:

make the simulation look like available historical data.

The goal is:

improve justified correspondence without destroying the simulation's ability to explore materially plausible reality beyond the observed dataset.

Minimum Implementation Framework

1. Receive the Governed Deviation

Use the SDMM deviation record as the alignment basis. 2. Determine Alignment Necessity

Establish whether the deviation requires action. 3. Assess Alignment Eligibility

Determine whether the available evidence is sufficient to justify modification. 4. Identify the Alignment Target

Determine which simulation element should change. 5. Define Alignment Scope

Bound the required modification. 6. Prioritize Alignment

Determine urgency and materiality. 7. Map Dependencies

Identify potentially affected simulation elements and downstream relationships. 8. Select the Alignment Pathway

Determine whether the response requires: calibration, correction, refinement, upstream lifecycle re-entry, scenario expansion, boundary restriction, or another controlled action.

9. Authorize and Implement Alignment

Apply the change under controlled configuration governance. 10. Re-Execute the Simulation

Produce a new governed execution basis. 11. Perform Post-Alignment Comparison

Compare the affected simulation behavior with the relevant Operational Reality basis. 12. Assess Alignment Effectiveness

Determine whether correspondence improved and whether new deviations emerged. 13. Preserve Residual Deviation and Uncertainty

Do not hide remaining differences. 14. Establish Alignment Status

Record the resulting bounded alignment state.

Governance Outputs

SRAM may produce: Alignment Requirement Record, Alignment Necessity Assessment, Alignment Eligibility Assessment, Alignment Priority, Alignment Target Record, Alignment Scope, Alignment Depth, Alignment Pathway, Alignment Dependency Map, Alignment Sequence, Alignment Authorization Record, Model Alignment Requirement, Environment Alignment Requirement, Parameter Alignment Requirement, Assumption Alignment Requirement, Input Alignment Requirement, Scenario Alignment Requirement, Behavioral Alignment Requirement, Temporal Alignment Requirement, Spatial Alignment Requirement, Interaction Alignment Requirement, Alignment Implementation Record, Alignment Version Record, Post-Alignment Comparison Record, Residual Deviation Record, Alignment Effectiveness Assessment, Alignment Conflict Record, Alignment Regression Record, Alignment Rollback Record, Reality Alignment Status, and Governed Reality Alignment Record.

Use Case 1 — Autonomous Robot Reality Gap

A warehouse robot performs reliably in simulation but shows materially longer stopping distances on real warehouse floors when:

payload is high + surface friction is reduced + rapid turning occurs.

SDMM identifies a conditional physical-behavior deviation.

SRAM determines that the appropriate Alignment Target is not the autonomous decision logic itself but the simulated physical interaction between:

payload + tire behavior + floor friction.

The environment and physical model are updated, the affected scenarios are re-executed, and post-alignment comparison determines whether the Reality Gap has materially decreased. Use Case 2 — Spacecraft Thermal Simulation

A spacecraft thermal simulation correlates strongly with ground tests and nominal mission telemetry but consistently underestimates temperature accumulation during a particular combination of:

high solar exposure + constrained orientation + reduced thermal dissipation.

SRAM classifies the deviation as requiring targeted alignment.

Instead of globally recalibrating the entire thermal model, the affected environmental and subsystem relationship is modified and the relevant mission scenarios are re-executed.

The updated simulation improves correspondence in the affected region while preserving separate uncertainty for extreme mission conditions that have never been directly observed.

Architectural Position

SRAM is the fourth module of the Simulation-to-Reality Correlation Standard (SRCS).

The progression is:

SRCM — Simulation-Reality Comparison

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SCCM — Simulation Correlation Criteria

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SDMM — Simulation-Reality Deviation

↓

SRAM — Reality Alignment

↓

SCUM — Correlation Uncertainty

SRAM occupies the point where:

knowledge of simulation-reality difference becomes governed action toward improved correspondence.

Foundational Principle

Reality Alignment must reduce unjustified simulation-reality difference without converting simulation into an overfitted imitation of observed reality or implying that simulation and reality have become equivalent.

Canonical Closing Statement

Reality Alignment Module (SRAM) governs how identified Simulation-Reality Deviations are translated into justified, bounded, traceable, and reassessable alignment actions. By separating deviation from alignment necessity, observed deviation location from alignment target, modification from demonstrated alignment, parameter fitting from reality correspondence, and alignment from reality equivalence, SRAM enables simulations to improve against Operational Reality without erasing legitimate abstraction, uncertainty, or unobserved but materially plausible conditions. Through controlled alignment scope, dependency mapping, lifecycle re-entry, re-execution, post-alignment comparison, residual-deviation preservation, regression detection, and bounded alignment status, SRAM establishes the governed pathway through which simulation models may learn from reality while remaining simulations rather than becoming artificial reproductions of historical observations.

Canonical Source

https://originopenfoundation.org/